



DIGITAL OPERATIONS & MAINTENANCE PLATFORM

Stakeholder Deployment & Operations Guide

An institutional operations platform unifying student residents, on-site maintenance technicians, and university administration into a closed-loop digital ecosystem.

STUDENT PORTAL

Resident Experience

Password-free room verification, instant fault reporting, photo evidence & real-time ticket tracking.

corridor-hills-residence.vercel.app/report

MAINTENANCE STAFF

Technician Terminal

Mobile-first field dispatch, live work timers, skill backlog queues, parts pause & proof-of-work uploads.

corridor-hills-residence.vercel.app/staff/login

ADMINISTRATION

Operations Control

Operations KPIs, live SLA radar, digital twin unit mapping, staff roster, CSV imports & audit logging.

corridor-hills-residence.vercel.app/admin/login

EXECUTIVE BRIEFING FOR TUT OPERATIONS & FACILITIES LEADERSHIP

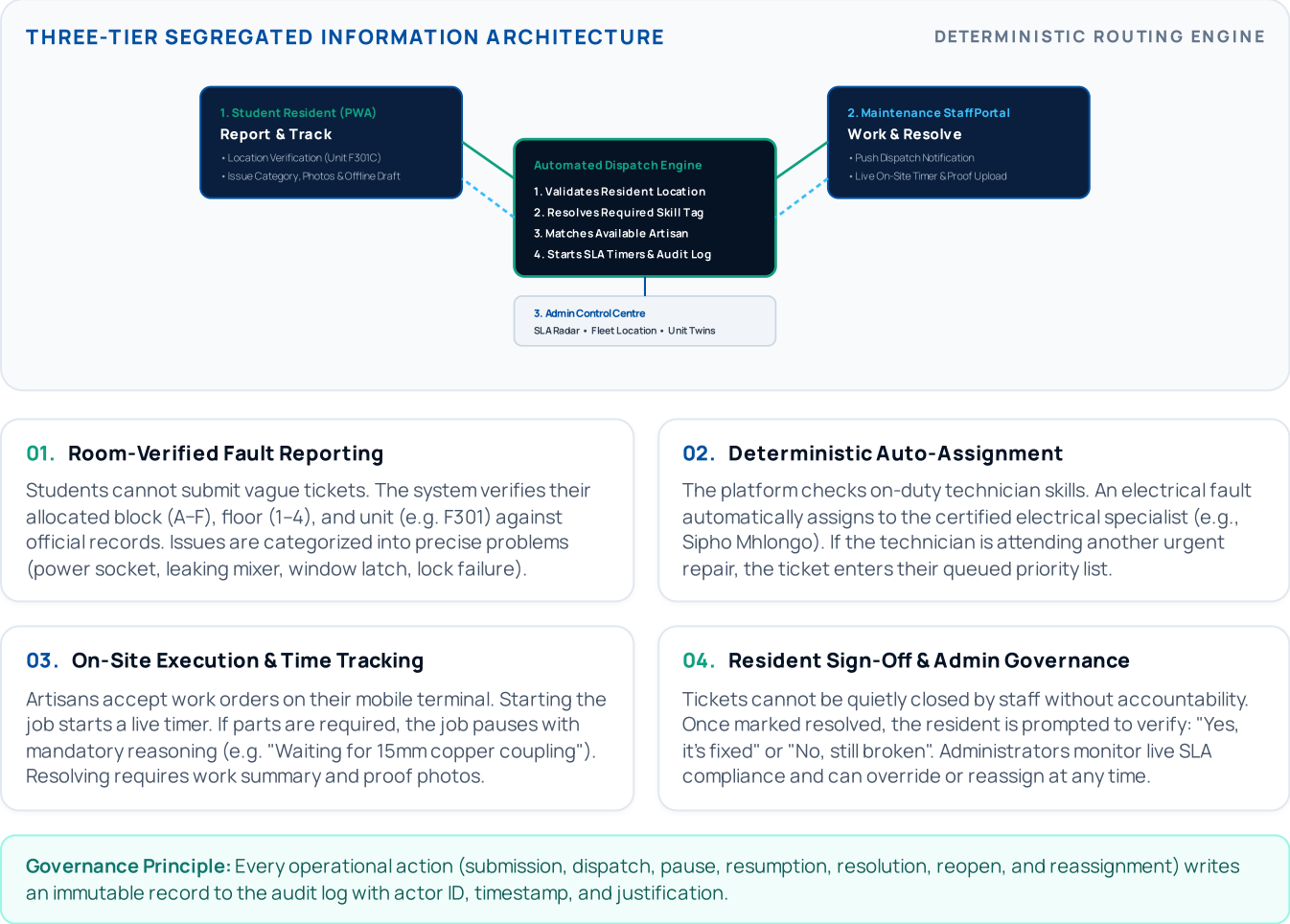
This document provides the operational blueprint, system architecture, role-based access protocols, and pre-configured demo datasets required to evaluate the live Corridor Hills Residence platform. The system is deployed and accessible on any desktop or mobile browser without requiring app store installations.



OPERATIONAL BLUEPRINT

How the System Works: End-to-End Workflow

The Corridor Hills platform eliminates disconnected spreadsheets and manual logbooks by orchestrating issues across three specialized interfaces tied to a deterministic auto-dispatch engine.



STUDENT EXPERIENCE • CAMPUS HORIZON

Report → Track → Resolve

A frictionless, password-free student interface optimized for mobile devices, enabling students to report issues in under 60 seconds with offline safety.

HOW STUDENTS ACCESS & REPORT

- 1 Navigate to Resident Portal**
Visit /report or /login on any smartphone or laptop. No password or app store download required.
- 2 Confirm Room Allocation**
Select Unit (e.g. F301), Bedroom (A, B, or C), and enter official 9-digit TUT student number.
- 3 Step Through Fault Wizard**
Select Area (Room/Kitchen/Bath) → Category (Electrical/Plumbing/Locks) → Specific Issue → Take/Attach Photo → Submit.
- 4 Track Live Progress**
View technician assignment, arrival status, live status timeline, and confirm resolution upon repair.

DEMO RESIDENT TESTING DATA

Stakeholders can test resident reporting with the following pre-registered residence allocations:

FIELD	RECOMMENDED DEMO VALUE
Residence Unit	F301 (Block F, Floor 3, Unit 01)
Allocated Bedroom	Room C (Full Location: F301C)
TUT Student #	220123456
Alternative Test 1	Unit A101 • Room A • Student 221849201
Alternative Test 2	Unit E204 • Room B • Student 219483029
1-Tap Shortcut	Click "Use sample resident (F301C)" on the login screen to populate all fields with one click.

Campus Scenery Customization

Residents can tap the camera icon in the bottom corner or header to switch their backdrop between **Evening Glow** (sunset), **After the Rain** (rainbow), and **After Hours** (night lighting). The platform remembers their choice locally.

STAFF PORTAL • 24/7 OPERATIONS

Receive → Prioritise → Work → Resolve

A high-contrast, mobile-first interface engineered for on-duty technicians operating under direct sunlight or with gloves, featuring touch targets $\geq 48\text{px}$ and offline job caching.

TECHNICIAN WORK ORDER STATE MACHINE

ENFORCED VERIFICATION PATH



DEMO STAFF ROSTER (1-TAP LOGIN)

Visit `/staff/login` to select any active technician:

STAFF ID	ARTISAN NAME	SPECIALTY	CURRENT STATUS
CH-ST-001	Sipho Mhlongo	Electrical Specialist	AVAILABLE
CH-ST-002	David Khumalo	Plumbing Specialist	BUSY
CH-ST-003	Thabo Ndlovu	Carpentry & Access	AVAILABLE
CH-ST-004	Nomsa Zulu	Senior Facilities (HVAC)	ON BREAK

KEY FEATURES TO TEST ON STAFF PORTAL

A

Current Job Card & Live Timer

Accept an assigned work order and press "Start Work". A persistent on-site timer records active labor duration.

B

Pause Work (Parts / Access Required)

Select "Pause Work" and choose a reason: Parts Needed, Access Denied, or Specialist Required. Halts SLA timer with an audit note.

C

Unassigned Backlog Queue (`/staff/queue`)

Technicians can browse unassigned tickets filtered by their certified trade skills and claim jobs directly.



OPERATIONS CONTROL CENTRE

Administration Portal & 5-Minute Testing Guide

Supervisors and facilities leadership access live facility intelligence, reassign technicians with mandatory justifications, and track SLA health across all 6 blocks.

ADMIN ACCOUNTS (/ADMIN/LOGIN)

ADMIN ID	NAME & ROLE	PERMISSIONS
CH-ADM-001	Lerato Molefe Operations Lead	SUPER-ADMIN Full Reassign, Staff, Units, Audit
CH-ADM-002	Katlego Dlamini Facilities Supervisor	DISPATCH Dispatch, Reassign, SLA Watch, Reports

KEY ADMIN MODULES

- **SLA Watch Radar:** Real-time countdowns tracking Emergency (≤2h), Urgent (≤12h), and Routine (≤48h) tickets.
- **Digital Twin Units (/admin/units):** Inspect Block A-F health and repeat fault history.
- **Immutable Audit Log (/admin/audit):** Complete forensic trail of every single user event.
- **Resident Directory (/admin/residents):** Search and export/import CSV roster.

RECOMMENDED 5-MINUTE TEST SEQUENCE

- 1 **Report Issue as Student (1 Min)**

Go to /report, click "Use sample resident (F301C)", select **Electrical → Power Socket Fault**, and submit. Note the ticket reference.
- 2 **Execute as Technician (2 Mins)**

Go to /staff/login, tap Sipho Mhlongo (CH-ST-001). See the new job, click "Accept Work Order" → "Start Work" → "Complete Work" with summary.
- 3 **Confirm Resolution as Student (1 Min)**

Return to /requests. Click the ticket → tap "Yes, it's fixed" to close it, or "No, still broken" to test reopening escalation.
- 4 **Audit Trail in Admin Centre (1 Min)**

Go to /admin/login, tap Lerato Molefe (CH-ADM-001). View /admin/audit to see the full timestamped log of the job.